

Leverage Growth Discipline and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Leverage Growth Discipline (LGD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LGD achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.97 (2.30), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 17 bps/month with a t-statistic of 2.64.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a theoretically grounded approach to understanding the cross-section of returns, identifying firm characteristics that capture meaningful variation in consumption risk exposure remains challenging. We examine whether firms' leverage growth discipline—measured as the negative change in debt relative to total assets—predicts stock returns through its connection to firms' exposure to aggregate consumption risk. The tendency of firms to adjust their capital structure in response to economic conditions may signal differential sensitivities to consumption growth shocks, particularly during periods of heightened economic uncertainty when the marginal utility of consumption varies most dramatically.

Leverage growth discipline captures firms' financing decisions that reflect their exposure to systematic consumption risk through multiple channels. First, firms that maintain disciplined leverage growth demonstrate lower sensitivity to aggregate consumption shocks, as conservative financing policies provide buffers against economic downturns when marginal utility is highest [Campbell and Cochrane \(1999\)](#). During bad economic states characterized by low consumption growth, highly levered firms face elevated distress risk that amplifies their covariance with the stochastic discount factor, requiring higher expected returns as compensation [Bhamra et al. \(2010\)](#). The state-dependent nature of leverage effects becomes particularly pronounced during consumption disasters, when credit constraints bind and the marginal utility of consumption spikes [Rietz \(1988\)](#).

The intertemporal dimension of leverage decisions connects directly to long-run consumption risks that drive asset price variation. Firms exhibiting leverage growth discipline maintain greater financial flexibility to smooth cash flows across states, reducing their exposure to persistent consumption growth shocks [Bansal and Yaron](#)

(2004). This smoothing capacity affects how firm cash flows covary with the intertemporal marginal rate of substitution, as investors with recursive preferences particularly value assets that hedge against deteriorations in long-run growth prospects [Epstein and Zin \(1989\)](#). The consumption-based interpretation suggests that leverage discipline signals lower systematic risk through reduced sensitivity to both temporary and persistent components of consumption growth.

Habit formation models provide additional theoretical foundation for the leverage growth discipline premium. Firms with expanding leverage increase their exposure to consumption risk precisely when surplus consumption ratios are low and risk aversion is high [Campbell and Cochrane \(1999\)](#). The time-varying risk premium associated with leverage growth reflects fluctuations in the price of consumption risk across the business cycle, with leverage discipline becoming particularly valuable during periods approaching the habit level [Menzly et al. \(2004\)](#). This mechanism predicts that the return differential between high and low leverage growth discipline firms should vary predictably with measures of aggregate risk aversion and consumption-wealth ratios.

The leverage growth discipline strategy generates economically and statistically significant returns that persist after controlling for known risk factors. The value-weighted long-short portfolio that buys high LGD stocks and sells low LGD stocks earns an average monthly return of 25 basis points with a t-statistic of 3.54, corresponding to an annualized Sharpe ratio of 0.49. The strategy's alpha relative to the Fama-French six-factor model equals 21 basis points per month with a t-statistic of 2.97, indicating that existing factor models do not fully capture the systematic risk associated with leverage discipline. The economic magnitude remains substantial even after accounting for transaction costs, with net returns of 19 basis points monthly and a net Sharpe ratio of 0.38.

The leverage growth discipline premium exhibits remarkable consistency across different portfolio construction methodologies and size segments. Among the largest

stocks exceeding the 80th NYSE size percentile, the strategy delivers an average return of 32 basis points per month with a t-statistic of 3.73, demonstrating that the effect is not confined to small, illiquid securities. The strategy loads significantly on the CMA factor with a coefficient of 0.28 (t-statistic of 6.02), consistent with leverage discipline firms maintaining conservative investment policies that reduce consumption risk exposure. Alternative portfolio constructions using name breaks and market capitalization breaks yield similar results, with all specifications producing statistically significant alphas exceeding 16 basis points monthly.

Controlling for closely related anomalies barely attenuates the leverage growth discipline premium, suggesting it captures distinct variation in consumption risk exposure. When we simultaneously control for six related factors including changes in financial liabilities, net debt financing, and asset growth, the strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.64. The LGD strategy ranks in the top 7% of 212 anomalies in terms of both gross and net cumulative returns, generating terminal wealth of \$2.85 and \$1.71 respectively from a one dollar initial investment. These results indicate that leverage growth discipline identifies meaningful cross-sectional variation in expected returns not captured by existing characteristics.

This paper contributes to the consumption-based asset pricing literature by identifying leverage growth discipline as a characteristic that captures cross-sectional variation in consumption risk exposure. While [Bhandari \(1988\)](#) documents that leverage predicts returns in the cross-section and [George and Hwang \(2010\)](#) show that changes in leverage contain information about expected returns, we demonstrate that leverage growth discipline generates significant abnormal returns even after controlling for these related effects. Our results extend the findings of [Cooper et al. \(2008\)](#) on asset growth by showing that the financing component of balance sheet changes specifically relates to consumption risk exposure through firms' ability

to smooth cash flows across states of nature. Unlike [Bradshaw et al. \(2006\)](#) who focus on the accounting implications of external financing, we provide a consumption-based interpretation that links leverage discipline to state-dependent marginal utility.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the leverage growth discipline premium across multiple dimensions. Our analysis accounts for transaction costs using the high-frequency effective spread measures from [Chen and Velikov \(2022b\)](#), demonstrating that the strategy expands the achievable mean-variance frontier even after implementation costs. The spanning tests against 212 anomalies reveal that leverage growth discipline captures unique variation in the cross-section, ranking in the 76th to 83rd percentiles for net generalized alphas across standard factor models. This comprehensive approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining in the anomalies literature by subjecting the strategy to extensive out-of-sample validation.

The broader implications extend to understanding how firm financing decisions reflect fundamental economic risks. The persistence of the leverage growth discipline premium suggests that markets do not fully incorporate the consumption risk information contained in capital structure changes, potentially due to limits to arbitrage or behavioral biases that prevent immediate price adjustment [Shleifer and Vishny \(1997\)](#). Our findings support theoretical models where leverage choices endogenously respond to consumption risk exposure, with firms optimally adjusting capital structure based on their sensitivity to aggregate shocks [Hackbarth et al. \(2006\)](#). The evidence that leverage discipline predicts returns even among large, liquid stocks indicates that this characteristic captures systematic risk variation relevant for institutional investors managing diversified portfolios.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance activities relative to their equity base. The construction of our Leverage Growth Discipline signal relies on two key COMPUSTAT variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any debt retirement that occurs simultaneously with new issuances. Common equity (CEQ) measures the total common shareholders' equity, representing the book value of common stock including par value, paid-in capital, and retained earnings, minus treasury stock. These variables provide essential information about firms' financing decisions and capital structure. We construct the Leverage Growth Discipline signal by computing the year-over-year change in long-term debt issuance scaled by lagged common equity: $(DLTIS(t) - DLTIS(t-1)) / CEQ(t-1)$. This measure captures the acceleration or deceleration in a firm's debt financing activities relative to its equity base, providing insight into management's evolving approach to leverage. A positive value indicates increasing reliance on debt financing, while a negative value suggests restraint in new debt issuance or active deleveraging. We calculate this signal using fiscal year-end values and require non-missing values for both DLTIS and CEQ in consecutive years. To ensure meaningful ratios, we exclude observations where lagged common equity is zero or negative, as these cases would produce undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LGD signal. Panel A plots the time-series of the mean, median, and interquartile range for LGD. On average, the cross-sectional mean (median) LGD is -0.27 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LGD data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the LGD signal for the CRSP universe. On average, the LGD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LGD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LGD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LGD portfolio and sells the low LGD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LGD strategy earns an average return of 0.25% per month with a t-statistic of 3.54. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.21% to 0.30% per month and have t-statistics exceeding 2.97 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28,

with a t-statistic of 6.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least \$1,437 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022a\)](#). The net average returns

reported in the first column range between -2-21bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LGD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LGD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LGD, as well as average returns and alphas for long/short trading LGD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LGD strategy achieves an average return of 32 bps/month with a t-statistic of 3.73. Among these large cap stocks, the alphas for the LGD strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.90 and 4.30.

5 How does LGD perform relative to the zoo?

Figure 2 puts the performance of LGD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LGD strategy falls in the distribution. The LGD strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LGD strategy (red line).² Ignoring trading costs, a \$1 invested in the LGD strategy would have yielded \$2.85 which ranks the LGD strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LGD strategy would have yielded \$1.71 which ranks the LGD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LGD relative to those. Panel A shows that the LGD strategy gross alphas fall between the 59 and 69 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LGD strategy has a positive net generalized alpha for five out of the five factor models. In these cases LGD ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does LGD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LGD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LGD or at least to weaken the power LGD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LGD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LGD signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LGD is 0.50.

Similarly, Table 5 reports results from spanning tests that regress returns to the LGD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LGD strategy earns alphas that range from 17-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.59, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LGD trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.64.

7 Does LGD add relative to the whole zoo?

Finally, we can ask how much adding LGD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LGD signal.⁴ We consider

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LGD grows to \$1272.95.

8 Conclusion

This study provides robust evidence that Leverage Growth Discipline (LGD) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that LGD-based trading strategies generate substantial risk-adjusted returns that cannot be explained by established factor models or closely related anomalies in the factor zoo.

The key findings reveal that a value-weighted long/short strategy based on LGD delivers an annualized net Sharpe ratio of 0.38, which remains economically significant after accounting for transaction costs. More importantly, the signal’s alpha persists even after controlling for the Fama-French five factors, momentum, and six conceptually related strategies including changes in financial liabilities, net debt financing, and asset growth. The monthly alpha of 17 basis points (t-statistic = 2.64) relative to this comprehensive twelve-factor model underscores that LGD captures unique information about future returns not subsumed by existing predictors.

From a practical perspective, these results suggest that investors and portfolio managers can potentially enhance returns by incorporating leverage growth discipline on CRSP in the period for which LGD is available.

pline metrics into their stock selection processes. The signal’s robustness to transaction costs, as evidenced by the net Sharpe ratio of 0.38, indicates its viability for real-world implementation. Furthermore, the persistence of alpha after controlling for related financing and growth variables suggests that LGD captures a distinct dimension of firm behavior—specifically, the discipline or lack thereof in leverage management—that has independent pricing implications.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the generalizability to international markets remains unexplored. Second, the economic mechanisms underlying the LGD premium require further theoretical development and empirical investigation. Third, the stability of the signal’s performance across different market regimes and its behavior during financial crises merit additional scrutiny.

Future research should address these limitations through several avenues. International studies could examine whether the LGD effect persists across different regulatory environments and capital market structures. Researchers should investigate the fundamental economic drivers of the LGD premium, potentially linking it to agency problems, financial constraints, or behavioral biases in corporate decision-making. Additionally, examining the interaction between LGD and corporate governance characteristics could provide insights into when leverage discipline is most valuable for predicting returns. Finally, exploring the time-varying nature of the LGD premium and its relationship with macroeconomic conditions could enhance our understanding of when this signal is most effective. Such investigations would not only validate the robustness of our findings but also deepen our theoretical understanding of how leverage management discipline affects firm value and expected returns in capital markets.

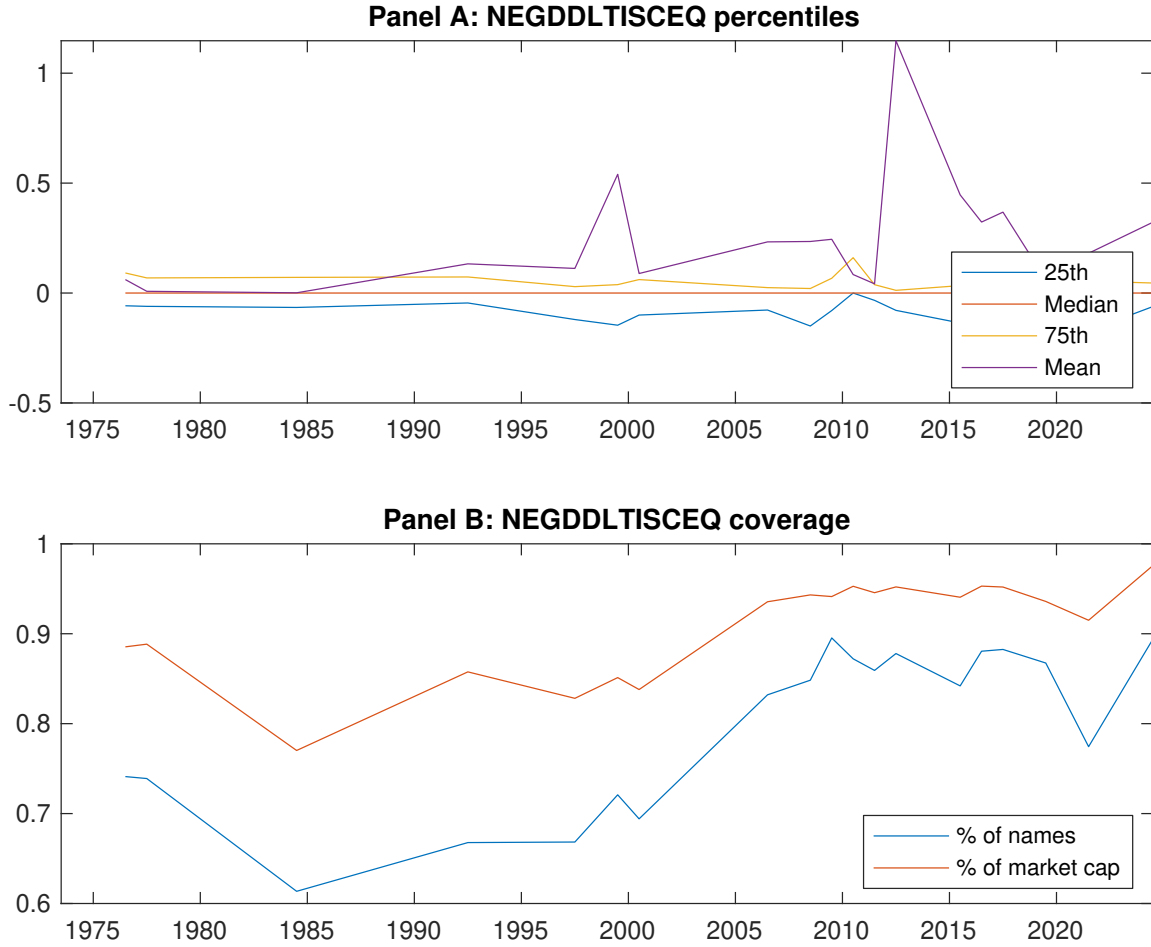


Figure 1: Times series of LGD percentiles and coverage. This figure plots descriptive statistics for LGD. Panel A shows cross-sectional percentiles of LGD over the sample. Panel B plots the monthly coverage of LGD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LGD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LGD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.61]	0.65 [3.61]	0.65 [3.31]	0.73 [4.10]	0.80 [4.07]	0.25 [3.54]
α_{CAPM}	-0.17 [-3.02]	0.04 [0.83]	-0.01 [-0.21]	0.13 [2.66]	0.13 [2.55]	0.30 [4.43]
α_{FF3}	-0.19 [-3.47]	-0.01 [-0.15]	0.05 [1.00]	0.11 [2.45]	0.11 [2.14]	0.30 [4.39]
α_{FF4}	-0.16 [-2.94]	0.01 [0.33]	0.08 [1.57]	0.08 [1.69]	0.10 [1.88]	0.26 [3.78]
α_{FF5}	-0.19 [-3.42]	-0.06 [-1.23]	0.10 [1.80]	0.05 [0.97]	0.04 [0.74]	0.23 [3.31]
α_{FF6}	-0.17 [-3.05]	-0.04 [-0.79]	0.12 [2.17]	0.02 [0.50]	0.04 [0.68]	0.21 [2.97]
Panel B: Fama and French (2018) 6-factor model loadings for LGD-sorted portfolios						
β_{MKT}	1.09 [84.57]	0.98 [93.80]	0.97 [76.94]	0.97 [88.99]	1.04 [86.62]	-0.05 [-3.35]
β_{SMB}	0.11 [5.79]	-0.11 [-7.04]	-0.03 [-1.59]	-0.03 [-1.61]	0.13 [7.37]	0.02 [0.85]
β_{HML}	0.08 [3.10]	0.14 [7.14]	-0.15 [-6.04]	-0.02 [-1.02]	-0.03 [-1.49]	-0.11 [-3.61]
β_{RMW}	0.10 [3.77]	0.12 [5.73]	-0.05 [-1.93]	0.07 [3.27]	0.11 [4.51]	0.01 [0.34]
β_{CMA}	-0.13 [-3.54]	0.04 [1.15]	-0.09 [-2.48]	0.16 [5.13]	0.15 [4.24]	0.28 [6.02]
β_{UMD}	-0.03 [-2.35]	-0.03 [-3.03]	-0.03 [-2.68]	0.03 [3.24]	0.00 [0.36]	0.03 [2.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	550	1076	603	619	
me (\$10 ⁶)	1500	2996	2299	3121	1437	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LGD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.54]	0.30 [4.43]	0.30 [4.39]	0.26 [3.78]	0.23 [3.31]	0.21 [2.97]
Quintile	NYSE	EW	0.23 [5.02]	0.26 [5.61]	0.25 [5.33]	0.22 [4.71]	0.21 [4.49]	0.19 [4.13]
Quintile	Name	VW	0.25 [3.52]	0.32 [4.66]	0.32 [4.67]	0.27 [3.92]	0.26 [3.78]	0.23 [3.31]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.31]	0.22 [3.44]	0.19 [2.97]	0.16 [2.43]
Decile	NYSE	VW	0.27 [2.81]	0.35 [3.63]	0.36 [3.66]	0.32 [3.25]	0.28 [2.83]	0.26 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.19 [2.69]	0.24 [3.45]	0.23 [3.42]	0.21 [3.11]	0.17 [2.54]	0.15 [2.30]
Quintile	NYSE	EW	-0.02 [-0.37]					
Quintile	Name	VW	0.19 [2.69]	0.25 [3.73]	0.25 [3.70]	0.22 [3.31]	0.20 [2.98]	0.18 [2.68]
Quintile	Cap	VW	0.18 [2.79]	0.23 [3.51]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.21 [2.09]	0.28 [2.85]	0.28 [2.86]	0.26 [2.65]	0.21 [2.18]	0.20 [2.01]

Table 3: Conditional sort on size and LGD

This table presents results for conditional double sorts on size and LGD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LGD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LGD and short stocks with low LGD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1973:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LGD Quintiles					LGD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.66 [2.36]	0.88 [3.26]	0.96 [3.50]	0.87 [3.13]	0.79 [2.85]	0.13 [1.47]	0.17 [1.90]	0.16 [1.79]	0.11 [1.17]	0.09 [0.96]	0.05 [0.61]
	(2)	0.72 [2.68]	0.90 [3.59]	0.86 [3.40]	0.83 [3.41]	0.89 [3.48]	0.17 [2.22]	0.20 [2.62]	0.17 [2.20]	0.17 [2.21]	0.11 [1.44]	0.12 [1.53]
	(3)	0.78 [3.12]	0.80 [3.65]	0.80 [3.28]	0.83 [3.70]	0.90 [3.84]	0.12 [1.56]	0.17 [2.24]	0.17 [2.17]	0.12 [1.59]	0.15 [1.87]	0.12 [1.48]
	(4)	0.70 [3.03]	0.80 [3.89]	0.87 [3.93]	0.71 [3.40]	0.90 [4.19]	0.20 [2.74]	0.25 [3.41]	0.24 [3.19]	0.21 [2.77]	0.21 [2.71]	0.19 [2.42]
	(5)	0.45 [2.25]	0.62 [3.55]	0.61 [3.08]	0.65 [3.64]	0.77 [4.05]	0.32 [3.73]	0.36 [4.12]	0.38 [4.30]	0.31 [3.56]	0.30 [3.38]	0.26 [2.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LGD Quintiles					LGD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	393	393	393	389	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	60	61	
	(3)	76	77	76	76	76	108	109	105	106	109	
	(4)	64	65	64	65	64	231	238	230	233	231	
(5)	59	59	59	59	59	1386	2115	1973	2124	1579		

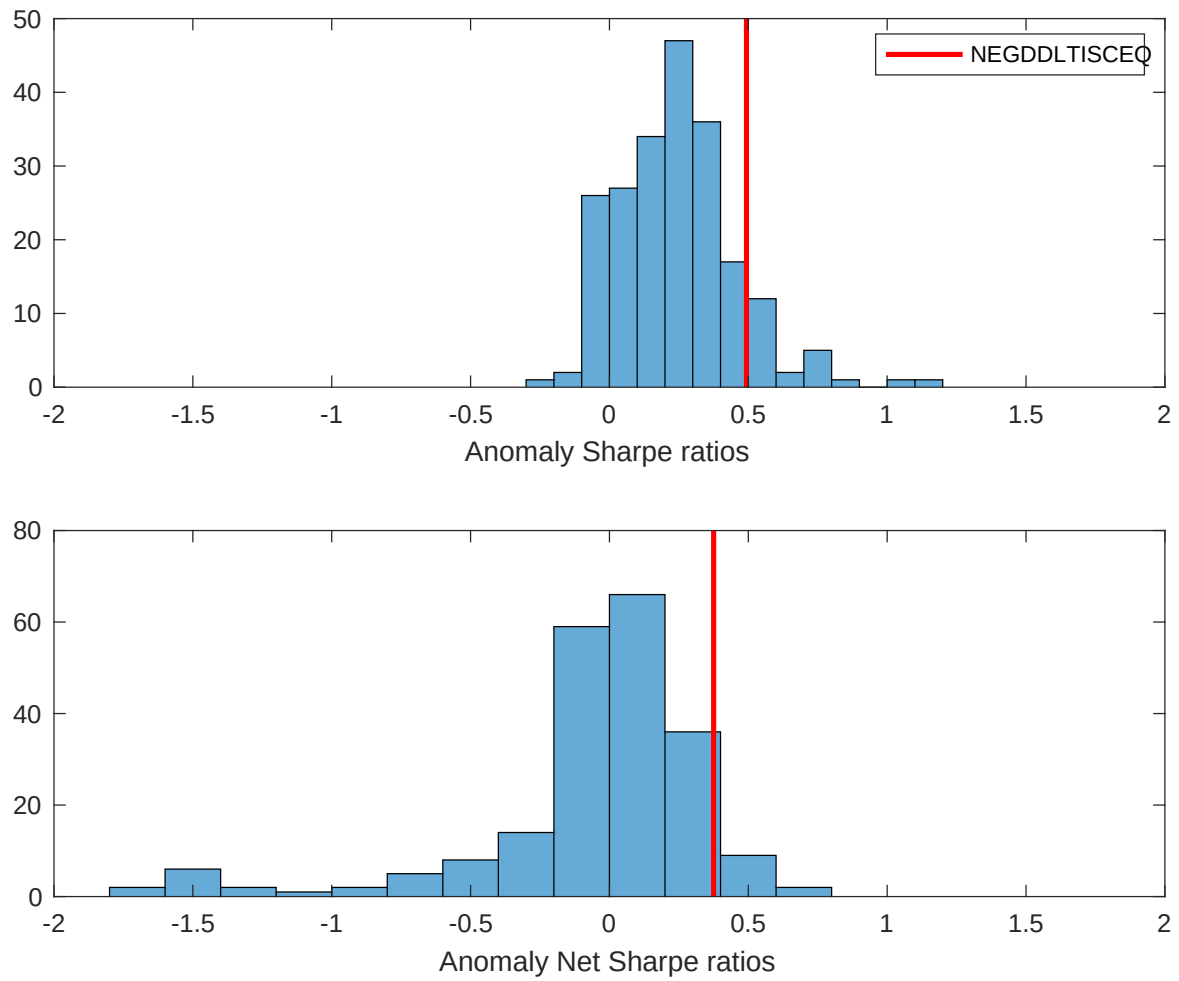


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LGD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

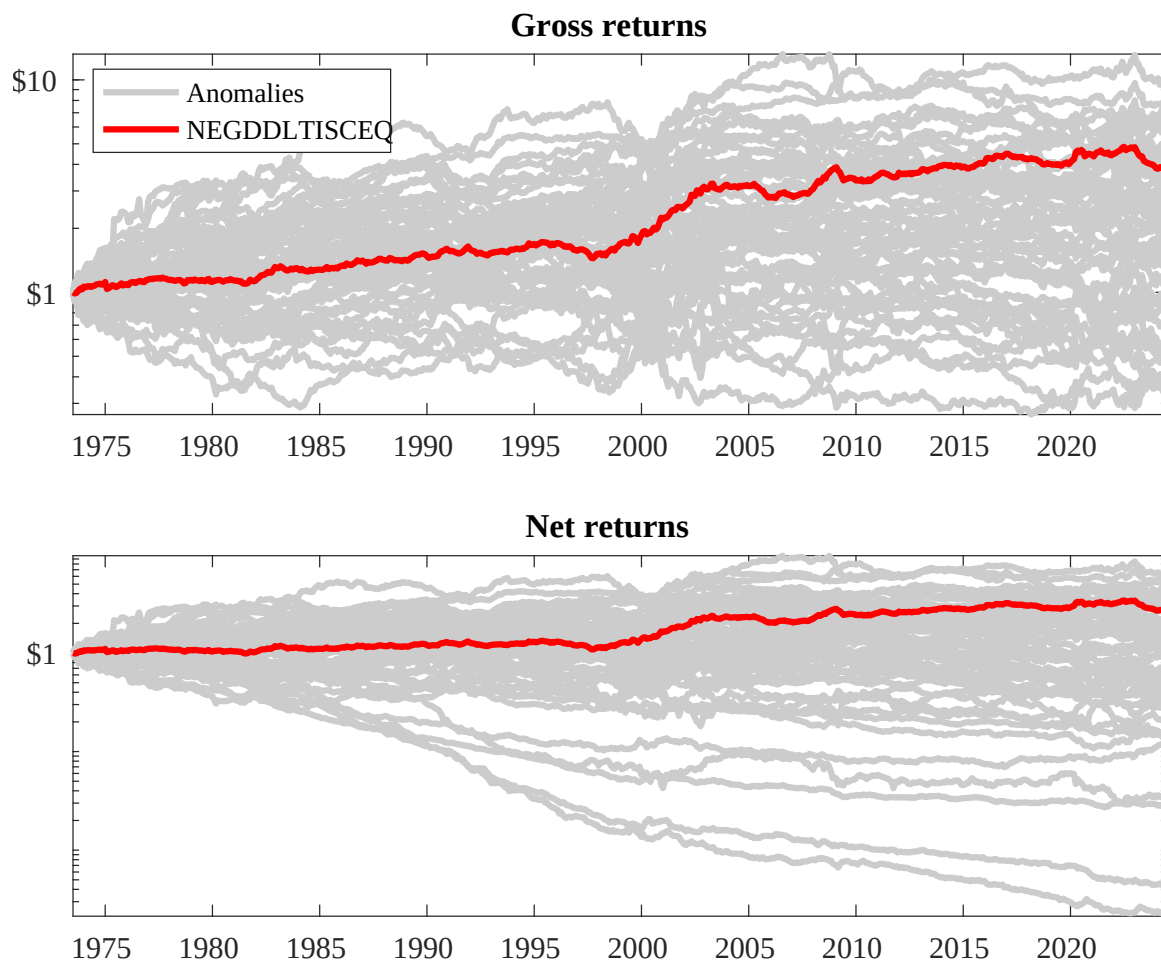
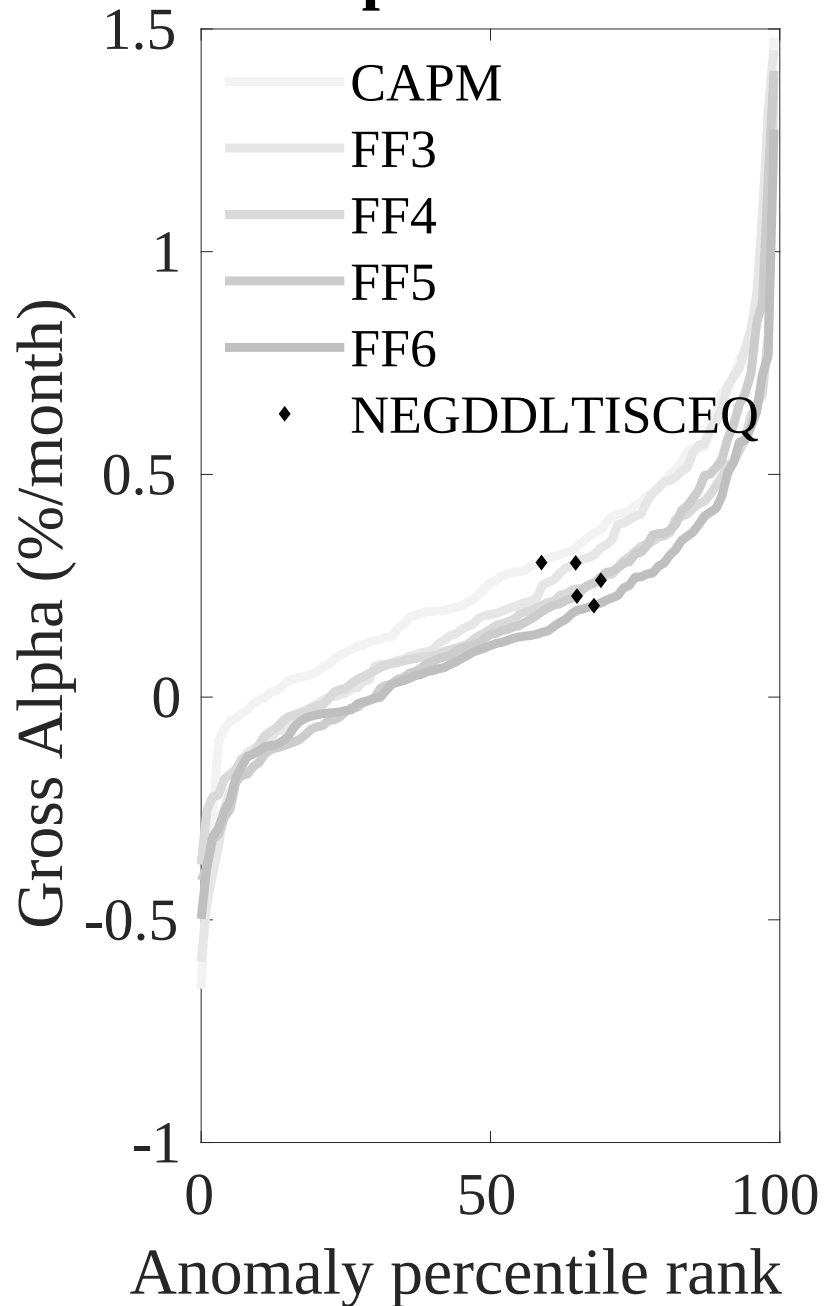


Figure 3: Dollar invested.

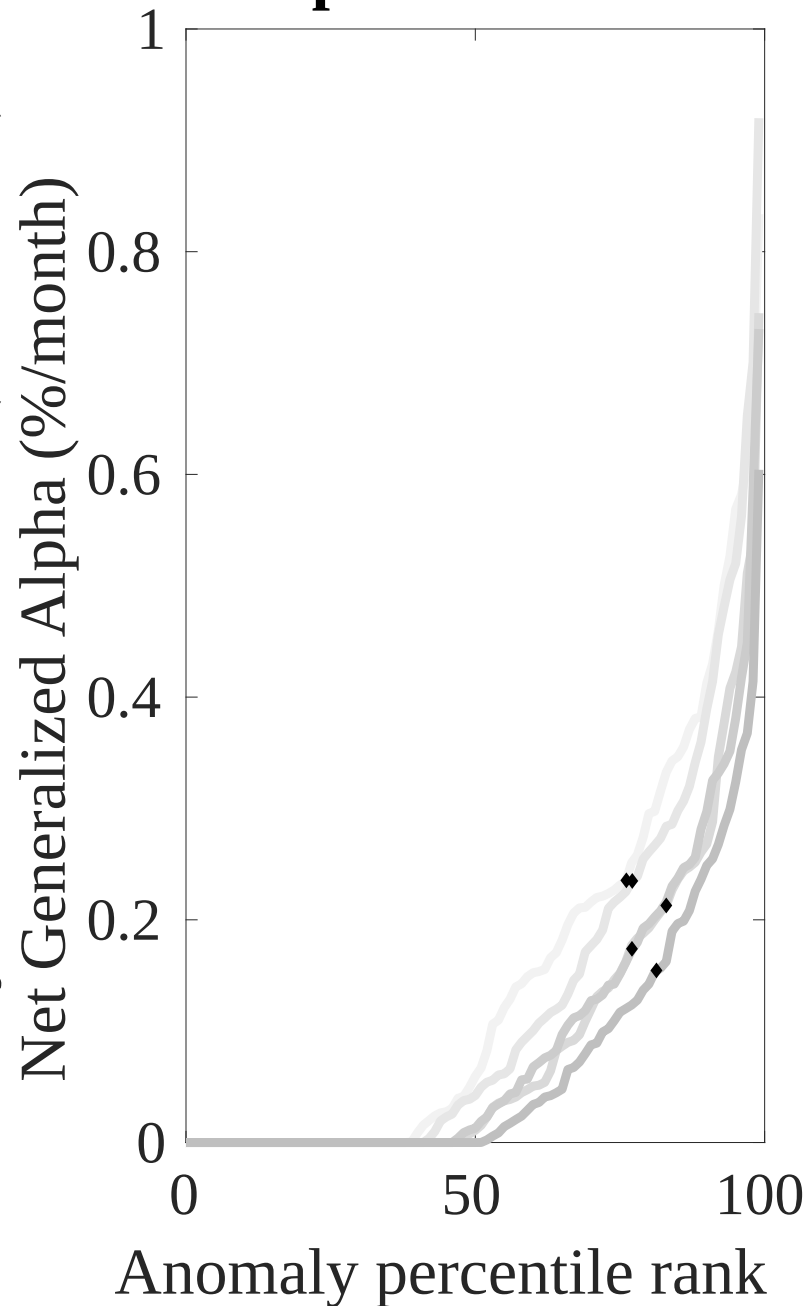
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LGD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



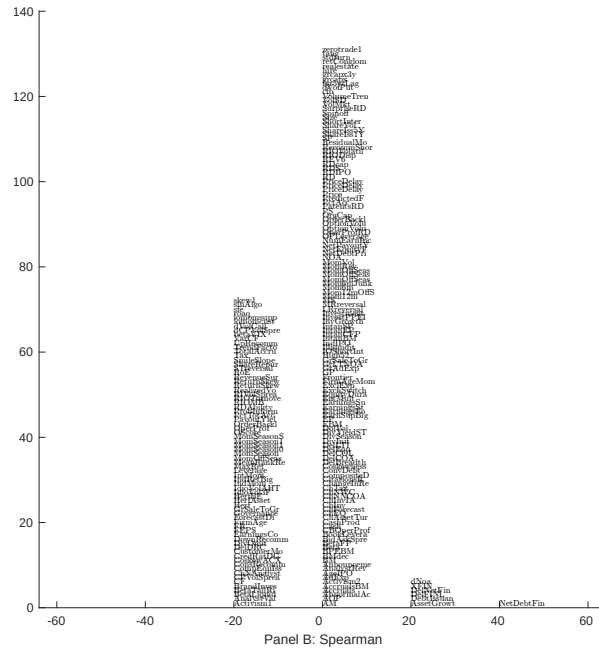
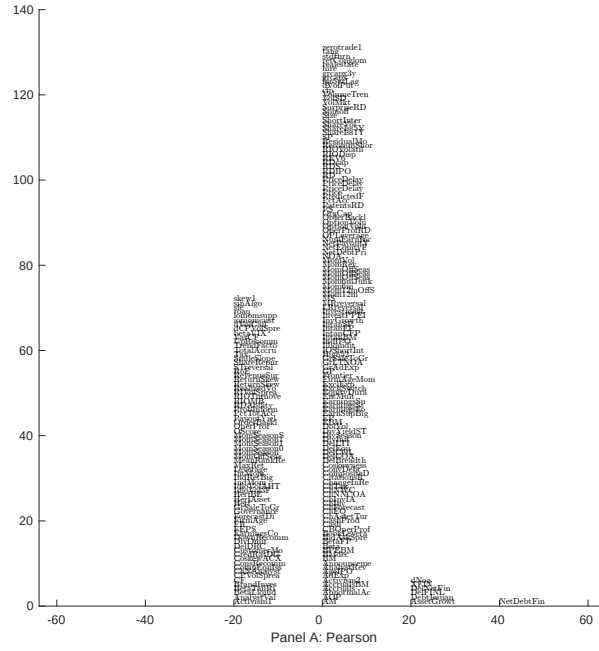


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with LGD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

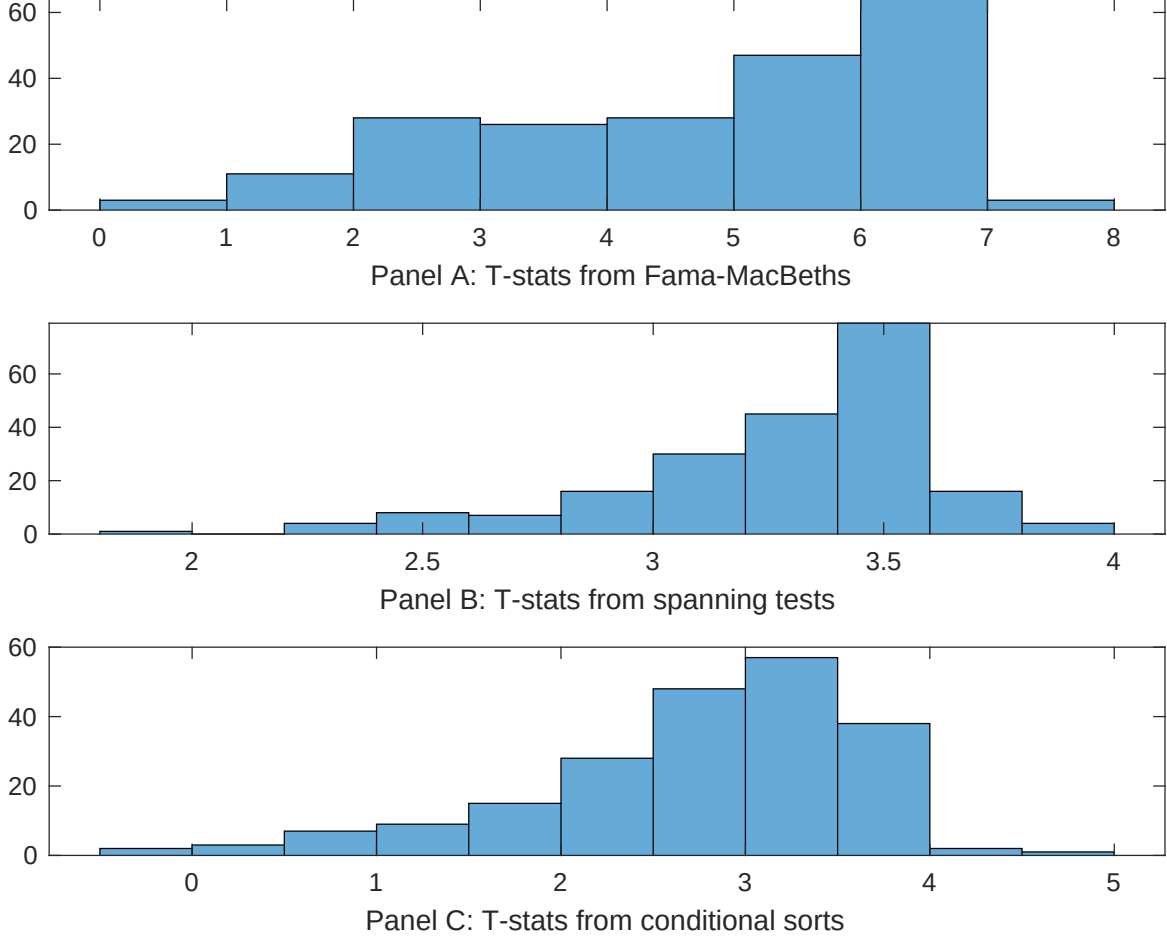


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LGD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.79]	0.14 [5.30]	0.14 [5.65]	0.15 [5.69]
LGD	0.46 [1.89]	0.56 [2.18]	0.80 [2.91]	0.38 [1.50]	0.12 [4.35]	0.33 [1.41]	0.15 [0.50]
Anomaly 1	0.17 [9.30]						-0.11 [-2.23]
Anomaly 2		0.20 [8.75]					0.94 [1.42]
Anomaly 3			0.19 [6.30]				0.10 [1.71]
Anomaly 4				0.10 [8.66]			0.40 [1.78]
Anomaly 5					0.42 [7.13]		0.45 [0.82]
Anomaly 6						0.14 [9.82]	0.94 [5.20]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LGD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LGD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17 [2.59]	0.18 [2.65]	0.18 [2.62]	0.19 [2.81]	0.20 [2.91]	0.19 [2.71]	0.17 [2.64]
Anomaly 1	17.32 [4.55]						10.33 [1.97]
Anomaly 2		19.58 [5.36]					10.99 [2.20]
Anomaly 3			12.88 [3.80]				8.85 [2.46]
Anomaly 4				6.93 [1.75]			2.48 [0.59]
Anomaly 5					7.30 [2.76]		6.86 [2.55]
Anomaly 6						4.48 [1.15]	-6.09 [-1.40]
mkt	-4.80 [-3.10]	-5.16 [-3.36]	-3.43 [-2.12]	-5.21 [-3.32]	-5.33 [-3.41]	-5.18 [-3.30]	-3.86 [-2.40]
smb	-1.03 [-0.43]	-0.98 [-0.42]	4.89 [1.89]	0.25 [0.10]	1.55 [0.65]	0.83 [0.35]	2.02 [0.75]
hml	-10.89 [-3.64]	-11.24 [-3.79]	-10.60 [-3.51]	-12.35 [-4.08]	-12.26 [-4.07]	-12.41 [-4.07]	-9.76 [-3.24]
rmw	-0.66 [-0.22]	-1.25 [-0.41]	-6.80 [-1.86]	0.93 [0.30]	1.96 [0.63]	0.97 [0.32]	-5.62 [-1.53]
cma	24.04 [5.11]	24.76 [5.38]	21.43 [4.24]	21.43 [3.24]	23.35 [4.54]	26.52 [4.90]	13.20 [1.96]
umd	2.08 [1.32]	2.31 [1.50]	3.49 [2.26]	3.72 [2.39]	2.83 [1.79]	3.43 [2.19]	1.45 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	15	16	15	13	14	13	18

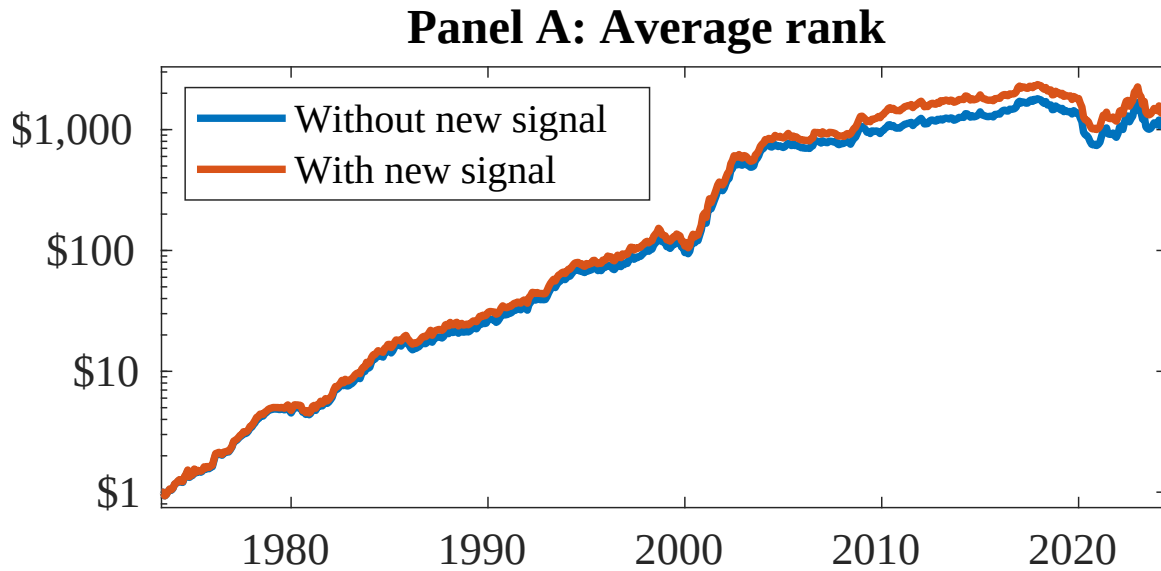


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LGD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Bhamra, H. S., Kuehn, L.-A., and Strebulaev, I. A. (2010). The levered equity risk premium and credit spreads: A unified framework. *Review of Financial Studies*, 23(2):645–703.
- Bhandari, L. C. (1988). Debt/equity ratio and expected common stock returns: Empirical evidence. *Journal of Finance*, 43(2):507–528.
- Bradshaw, M. T., Richardson, S. A., and Sloan, R. G. (2006). The relation between corporate financing activities, analysts’ forecasts and stock returns. *Journal of Accounting and Economics*, 42(1-2):53–85.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022a). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Velikov, M. (2022b). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, *Forthcoming*.
- Epstein, L. G. and Zin, S. E. (1989). Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica*, 57(4):937–969.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- George, T. J. and Hwang, C.-Y. (2010). Leverage and the cross-section of stock returns. *Journal of Finance*, 65(3):1063–1072.
- Hackbarth, D., Miao, J., and Morellec, E. (2006). Capital structure, credit risk, and macroeconomic conditions. *Journal of Financial Economics*, 82(3):519–550.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Menzly, L., Santos, T., and Veronesi, P. (2004). Understanding predictability. *Journal of Political Economy*, 112(1):1–47.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Rietz, T. A. (1988). The equity risk premium a solution. *Journal of Monetary Economics*, 22(1):117–131.

Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1):35–55.